T1

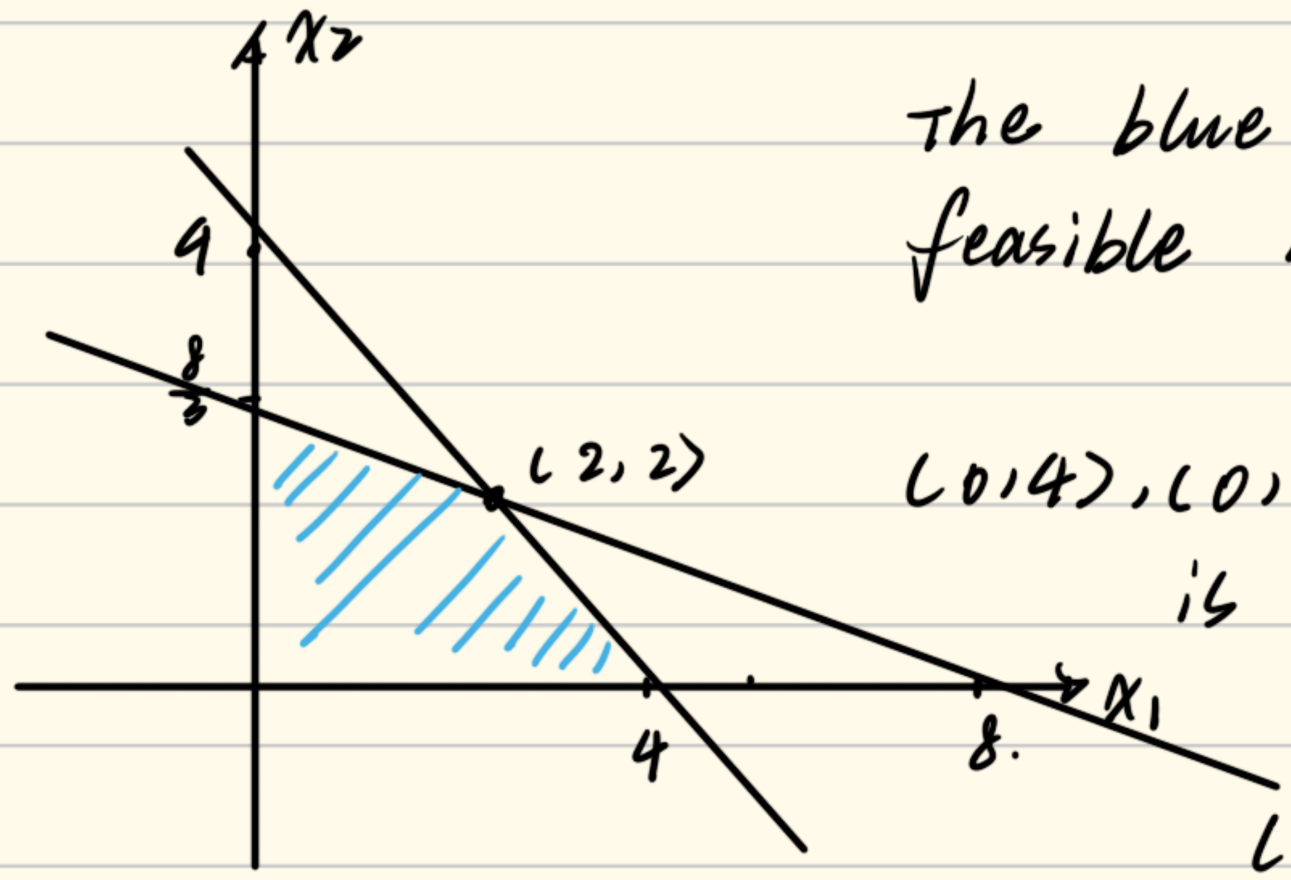
maximize $z = x_1 + 2x_2$

s.t $x_1 + 3x_2 \le 8$

$x_1 + x_2 \le 4$

$x_1 \ge 0, x_2 \ge 0.$

(a)



The blue region is the feasible region

$(0,4), (0,\frac{8}{3}), (0,0), (4,0), (8,0), (2,2)$ is all the corner-point solution, and $(0,\frac{8}{3})$, $(0,0), (4,0), (2,2)$ are feasible, $(0,4), (8,0)$ are unfeasible.

(b) $(0,\frac{8}{3})$: $x_1 = 0$, $x_2 = -\frac{1}{3}x_1 + \frac{8}{3}$

$(0,0)$: $x_1 = 0$, $x_2 = 0$.

$(4,0)$: $x_2 = 0$, $x_2 = -x_1 + 4$

$(2,2)$: $x_2 = -\frac{1}{3}x_1 + \frac{8}{3}$, $x_2 = -x_1 + 4$

(c) $\because$ $z = x_1 + 2x_2$ can be transformed like that:

$x_2 = -\frac{1}{2}x_1 + \frac{z}{2}$

consider $(0,\frac{8}{3})$, $z = \frac{16}{3}$

consider $(0,0)$, $z = 0$

consider $(4,0)$, $z = 4$

consider $(2,2)$, $z = 6$.

As a result, the optimal solution is $(2,2)$, which $z = 6$

(d) After introducing slack variables, the LP can be transformed like that:

$$\begin{cases} z - x_1 - 2x_2 = 0 \\ x_1 + 3x_2 + x_3 = 8 \\ x_1 + x_2 + x_4 = 4 \\ x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0. \end{cases}$$

(x_3, x_4 are slack variables).

(e) $(0, \frac{8}{3})$: $x_1 = 0, x_2 = \frac{8}{3}$

$\therefore x_3 = 8 - x_1 - 3x_2 = 8 - 0 - 8 = 0$

$x_4 = 4 - x_1 - x_2 = 4 - 0 - \frac{8}{3} = \frac{4}{3}$

$\therefore (0, \frac{8}{3}, 0, \frac{4}{3})$ is the BF solution.

$(0, 0)$: $x_1 = 0, x_2 = 0$

$\therefore x_3 = 8, x_4 = 4$

$\therefore (0, 0, 8, 4)$ is the BF solution.

$(4, 0)$: $x_1 = 4, x_2 = 0$

$\therefore x_3 = 4, x_4 = 0$

$\therefore (4, 0, 4, 0)$ is the BF solution.

$(2, 2)$: $x_1 = 2, x_2 = 2$

$\therefore x_3 = 0, x_4 = 0$

$\therefore (2, 2, 0, 0)$ is the BF solution.

(f): $(0, \frac{8}{3}, 0, \frac{4}{3})$: basic variables : x_2, x_4
non-basic. variables: x_1, x_3.

$(0, 0, 8, 4)$: basic variables : x_3, x_4
non-basic. variables: x_1, x_2.

$(4, 0, 4, 0)$: basic variables : x_1, x_3
non-basic. variables: x_2, x_4

$(2, 2, 0, 0)$: basic variables : x_1, x_2

non-basic. variables: x_3, x_4

T2.

$$\text{maximize} \quad z = 2x_1 + 3x_2$$

$$\text{s.t.} \quad -3x_1 + x_2 \le 1$$

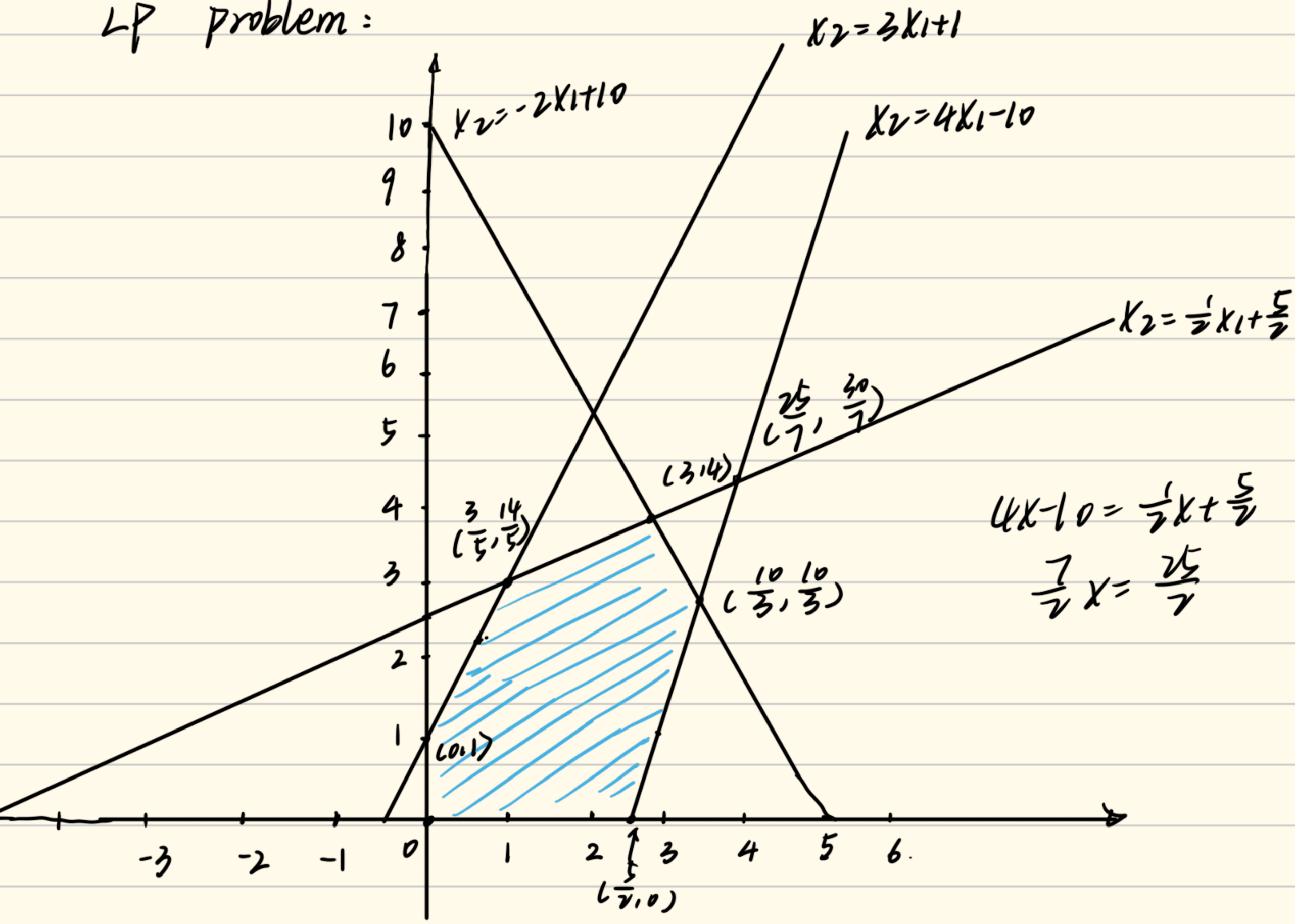
$$4x_1 + 2x_2 \le 20$$

$$4x_1 - x_2 \le 10$$

$$-x_1 + 2x_2 \le 5$$

$$x_1, x_2 \ge 0.$$

According to the title, we can graphically express this LP problem:



the blue region is the feasible region.

let $x_2 = \frac{1}{2}x_1 + \frac{5}{2} = 3x_1 + 1 \Rightarrow x_1 = \frac{3}{5}, x_2 = \frac{14}{5}$

let $\frac{1}{2}x_1 + \frac{5}{2} = -2x_1 + 10 \Rightarrow x_1 = 3, x_2 = 4$

let $-2x_1 + 10 = 4x_1 - 10 \Rightarrow x_1 = \frac{10}{3}, x_2 = \frac{10}{3}$

$(0,0), (\frac{5}{2},0), (0,1), (\frac{3}{5},\frac{14}{5}), (3,4), (\frac{10}{3},\frac{10}{3})$

are the feasible corner-point solution.

start at $(0,0)$:

$\because z = 2x_1 + 2x_2 \Rightarrow z(0,0) = 0$.

$(\frac{5}{2},0)$: $z(\frac{5}{2},0) = 5$

$(0,1)$: $z(0,1) = 3$.

$(\frac{3}{5},\frac{14}{5})$: $z(\frac{3}{5},\frac{14}{5}) = \frac{48}{5}$

$(3,4)$: $z(3,4) = 18$.

$(\frac{10}{3},\frac{10}{3})$: $z(\frac{10}{3},\frac{10}{3}) = \frac{50}{3}$

As a result, the optimal solution is $(3,4)$, which $z = 18$.

T3

maximize $z = x_1 + 2x_2 + 4x_3$

s.t. $3x_1 + x_2 + 5x_3 \le 10$

$x_1 + 4x_2 + x_3 \le 8$

$2x_1 + 2x_3 \le 7$

$x_1, x_2, x_3 \ge 0$.

Introduce the slack variables s_1, s_2, s_3 $(s_i \ge 0)$.

So the LP problem can be transformed into:

$z - x_1 - 2x_2 - 4x_3 = 0$

$3x_1 + x_2 + 5x_3 + s_1 = 10$

$x_1 + 4x_2 + x_3 + s_2 = 8$

$2x_1 + 2x_3 + s_3 = 7$

$x_1, x_2, x_3, s_1, s_2, s_3 \ge 0$.

Iteration	Basic variable	row	Coefficient							RHS	Ratio
			z	x_1	x_2	x_3	s_1	s_2	s_3		
	z	(0)	1	-1	-2	-4	0	0	0	0	
0	s_1	(1)	0	3	1	5	1	0	0	10	2

(0,0,0,0,0,8,7)	s_2	(2)	0	1	4	1	0	1	0	8	8
	s_3	(3)	0	2	0	2	0	0	1	7	3.5
1	z	(0)	1	$\frac{7}{5}$	$-\frac{6}{5}$	0	$\frac{4}{5}$	0	0	8	
	x_3	(1)	0	$\frac{3}{5}$	$\frac{1}{5}$	1	$\frac{1}{5}$	0	0	2	10
	s_2	(2)	0	$\frac{2}{5}$	$\frac{19}{5}$	0	$-\frac{1}{5}$	1	0	6	$\frac{30}{19}$
	s_3	(3)	0	$\frac{4}{5}$	$-\frac{2}{5}$	0	$-\frac{2}{5}$	0	1	3	$-\frac{15}{2}$
2	z	(0)	1	$\frac{145}{19}$	0	0	$\frac{70}{19}$	$\frac{30}{19}$	0	$\frac{940}{19}$	
	x_3	(1)	0	$\frac{11}{19}$	0	1	$\frac{4}{19}$	$-\frac{1}{19}$	0	$\frac{32}{19}$	
	x_2	(2)	0	$\frac{2}{19}$	1	0	$-\frac{1}{19}$	$\frac{5}{19}$	0	$\frac{30}{19}$	
	s_3	(3)	0	$\frac{16}{19}$	0	0	$-\frac{8}{19}$	$\frac{2}{19}$	1	$\frac{69}{19}$	

We can solve that when $x_1 = 0$, $x_2 = \frac{30}{19}$, $x_3 = \frac{32}{19}$,

$z = 0 + \frac{30}{19} \times 2 + \frac{32}{19} \times 4 = \frac{188}{19}$.